

HW02

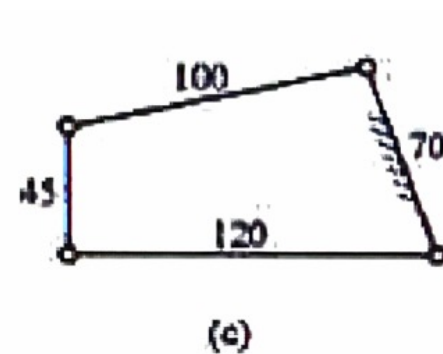
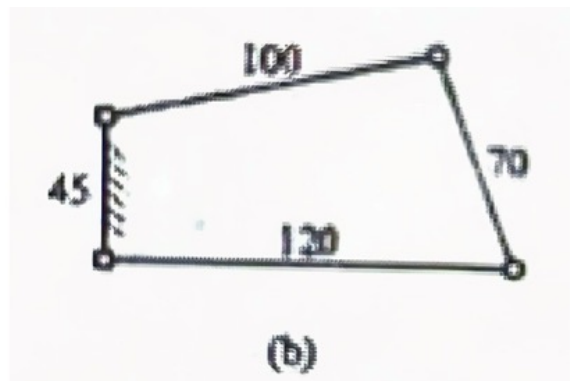
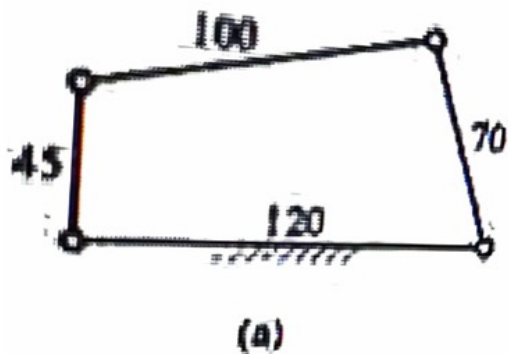
第02章 平面连杆机构 作业

所有作业要求手写

Autumn 2024

HW 02.1

- 根据长度关系判断图中三个铰链四杆机构：
图 (a) 是 _____ 机构、图 (b) 是 _____ 机构、图 (c) 是 _____ 机构。
- Based on the length relationships, identify the three four-bar linkage mechanisms shown in the figure:
Figure (a) is a _____ mechanism, Figure (b) is a _____ mechanism, and Figure (c) is a _____ mechanism.



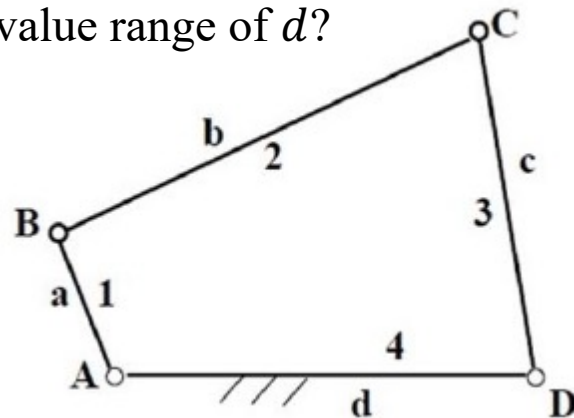
HW 02.2

• 如图所示，已知 $a = 240 \text{ mm}$, $b = 600 \text{ mm}$, $c = 400 \text{ mm}$, $d = 500 \text{ mm}$,

- 1) 当连杆 4 作为机架时，是否存在曲柄？
- 2) 是否可以选择不同的杆作为机架，以获得双曲柄机构和双摇杆机构？如何选择？
- 3) 若 a 、 b 、 c 三杆长度不变，取杆 4 为机架，为得到曲柄摇杆机构， d 的取值范是多少？

• As shown in the figure, it is known that $a = 240 \text{ mm}$, $b = 600 \text{ mm}$, $c = 400 \text{ mm}$, $d = 500 \text{ mm}$,

- 1) When the lever 4 is used as a frame, is there a crank?
- 2) Can I choose different bars for the frame to get the double crank mechanism and double rocker mechanism? How?
- 3) If the lengths of the three rods a , b , and c are unchanged, take rod 4 as the frame, and to obtain the crank and rocker mechanism, what should be the value range of d ?



HW 02.3

- 如图所示为偏置曲柄滑块机构。

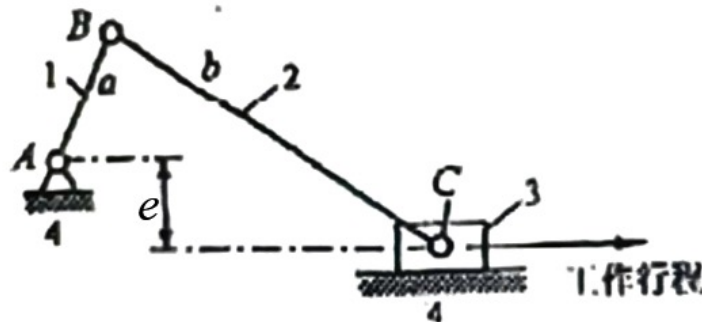
曲柄 1 和连杆 2 的长度分别为 201.7 mm 和 605.2 mm ，偏距 $e = 325.6\text{ mm}$ 。

- 1) 若滑块 3 向右运动为机构的工作行程（慢行程），试确定曲柄 1 的转动方向；
- 2) 求该机构的最小传动角 γ ，并作图画出其对应的位置。

- The figure shows an offset crank-slider mechanism.

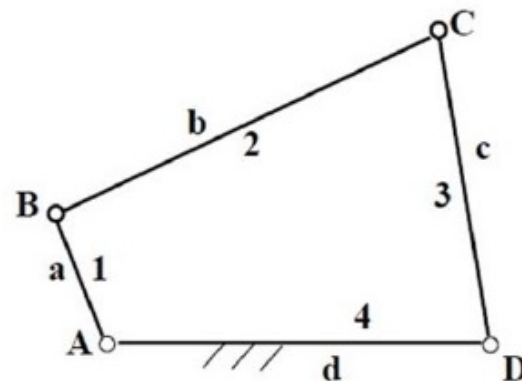
The lengths of crank 1 and connecting rod 2 are 201.7 mm and 605.2 mm , respectively, with an offset distance of $e = 325.6\text{ mm}$

- 1) If slider 3 moves to the right during the working stroke (slow stroke) of the mechanism, determine the rotation direction of crank 1.
- 2) Find the minimum transmission angle γ of the mechanism and draw its corresponding position.



HW 02.4

- 已知四杆机构 $L_1 = 28 \text{ mm}$, $L_2 = 52 \text{ mm}$, $L_3 = 50 \text{ mm}$, $L_4 = 72 \text{ mm}$, 当取 4 杆为机架时, 求该机构的极位夹角 θ 、最小传动角 γ 和行程速度变化系数 K 。
- It is known that the 4-pole mechanism $L_1 = 28 \text{ mm}$, $L_2 = 52 \text{ mm}$, $L_3 = 50 \text{ mm}$, $L_4 = 72 \text{ mm}$, when the 4-rod is taken as a rack, find crank angle θ between extreme positions, the minimum transmission angle γ and the coefficient of variation of travel speed of the mechanism K .



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Thank you~

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附加题 02.1

- 目前四足机器人腿部结构主要采用平面四连杆与串联两种构型，请问平面连杆机构在此的优势有哪些？
- At present, the leg structure of quadruped robot mainly adopts two configurations of plane four-bar linkage and series linkage. What are the advantages of the plane linkage mechanism in this condition?

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